

GPX Helper

One-page repo summary based on README and source files

WHAT IT IS

A local web app and Python toolkit for trimming GPX tracks to match video timestamps and for rendering GPX routes into MP4 map animations. The repo combines a Svelte frontend with a FastAPI backend.

WHO IT'S FOR

- GoPro and other action-camera users who need to sync footage with GPS tracks for mapping, overlays, or route exports.

WHAT IT DOES

- Trim a GPX file by a manual start and end time.
- Auto-fill trim times from embedded QuickTime video metadata in the browser.
- Check that requested trim windows fit inside the GPX time range.
- Split one GPX track into multiple clip-specific GPX files and download a ZIP.
- Estimate map-animation render time before starting the export.
- Render MP4 route animations with tile-style and line/marker controls.
- Expose health and capability endpoints plus file-download API responses.

HOW IT WORKS

- Frontend: Svelte + Vite UI (``frontend/src/App.svelte``) handles uploads, reads GPX timestamps, extracts MP4/MOV creation time from box metadata (``frontend/src/video-metadata.js``), and posts multipart forms.
- Backend: FastAPI (``backend/src/gpx_helper/api/main.py``) validates inputs, writes uploads to temp files, and streams GPX, ZIP, JSON ETA, or MP4 responses.
- GPX processing: ``gpx_splitter.py`` parses GPX XML, finds the closest timed track points, crops the track segment, and in CLI mode can read video timing via ``exiftool``.
- Map rendering: ``map_animator.py`` loads GPX points, converts them to Web Mercator, fetches map tiles, estimates render cost, and encodes MP4 output.
- Persistent database, auth, and deployment setup: Not found in repo.

HOW TO RUN

- Backend: ``cd backend && poetry install && poetry run uvicorn gpx_helper.api.main:app --reload``
- Frontend: ``cd frontend && npm install && npm run dev``
- Open the Vite URL (default ``http://localhost:4173``). The UI targets ``http://localhost:8000`` unless ``VITE_API_BASE`` is set.